A Formal Proof of Bertrand's Theorem in Lean 4

Nichlas Löfdahl Claude (Anthropic)*

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Abstract

Bertrand's theorem (1873) states that the only central forces for which bounded orbits are closed are the inverse-square law and Hooke's law. We give a machine-checked proof in Lean 4 with Mathlib of the local form of the theorem, in the spirit of Féjoz and Kaczmarek (after Herman). Suppose the potential V is smooth on an interval of radii, every circular orbit there is stable, and every orbit sufficiently close to a circular one is closed. Then $V'(r) = C/r^2$ or $V'(r) = Cr$ with $C > 0$. The apsidal angle is defined as an honest singular integral over the classically allowed region. The proof has two stages. First, a one-dimensional Morse lemma and a trigonometric substitution show that the apsidal angle is continuous in the energy up to the circular orbit, so closedness forces it to be constant. This makes $\sqrt{3 + rV''/V'}$ a rational constant, so the force is a power law. Second, in a smooth Morse chart the apsidal angle equals $\pi X'(0) + \frac{\pi}{2} X'''(0)\,\varepsilon + o(\varepsilon)$. Constancy forces $X'''(0) = 0$, which is the classical isochrony condition $5U'''^2 = 3U''U''''$, and for a power law this leaves only Newton and Hooke. Along the way we formalize a real Hadamard lemma and a smooth one-dimensional Morse lemma. To our knowledge this is the first formalization of Bertrand's theorem.

1 Introduction

Among all central forces, why do only gravity and the ideal spring produce closed orbits? Bertrand's answer [2] is a classic of mechanics, reproduced in Goldstein [5] and Arnold [1]. Rigorous treatments are surprisingly delicate. Arnold's sketch hides nontrivial analysis in its limits. Féjoz and Kaczmarek [4] give a careful local proof via Birkhoff invariants, and many elementary proofs have since appeared (e.g. [3]).

We formalize the theorem in two stages, both using only orbits near circular ones:

(I) *Power law.* If all orbits close to stable circular orbits are closed, the apsidal angle is constant near each circular orbit and the force is a power law $V' \propto r^{\beta^2 - 3}$ with rational β.

(II) *Newton or Hooke.* The second-order term of the apsidal angle in the energy must vanish. For a power law this leaves only $\beta = 1$ (Newton) and $\beta = 2$ (Hooke).

Stage (I) is the common first step of most proofs. For stage (II), many textbooks instead compare limits of the apsidal angle at extreme energies, which needs hypotheses on *all* bounded orbits. We follow the local route of Féjoz and Kaczmarek [4], which needs only near-circular orbits. Their abstract states the theorem in exactly this local form. Everything is checked in Lean 4 with Mathlib [6]. The link between closed trajectories of Newton's equations and the apsidal angle

*AI system (model Claude Opus 5.5), developed by Anthropic. Claude designed the formalization and proof strategy, wrote the Lean proofs (in part via parallel sub-agents) and drafted this manuscript in collaboration with the first author.

is taken as the definition of closedness (§9). The development is part of the repository `https://github.com/Crotalus/ai-formalized` (folder `bertrand`). It builds with Lean `v4.35.0` and Mathlib commit `56a0058`.

2 Setting

A particle of unit mass moves in a central potential $V(r)$ with angular momentum $M > 0$. In the variable $x = M/r$, Binet's equation states that the radial motion, parametrized by the polar angle θ, is a one-dimensional oscillation $\frac{1}{2}x'(\theta)^2 + U(x) = E$ in the effective potential

$$U(x) = \frac{x^2}{2} + V\!\left(\frac{M}{x}\right).$$

The *apsidal angle*, the angle from pericentre to apocentre, is the half-period of this oscillation:

$$\Phi(E) = \int_{\{U<E\}_{x_0}} \frac{dx}{\sqrt{2(E - U(x))}},$$

where $\{U < E\}_{x_0}$ is the connected component, containing the bottom x_0 of the well, of the classically allowed region. An orbit closes after finitely many revolutions exactly when $\Phi/\pi \in \mathbb{Q}$. A circular orbit of radius r has $M^2 = r^3 V'(r)$ and sits at the critical point $x_0 = M/r$ of U, with

$$U''(x_0) = 3 + \frac{rV''(r)}{V'(r)}, \tag{1}$$

and it is stable when this is positive. In Lean:

```
def apsidalAngle (U : ℝ → ℝ) (Ω : Set ℝ) (x₀ E : ℝ) : ℝ :=
  ∫ x in connectedComponentIn (Ω ∩ {x | U x < E}) x₀, 1 / √(2 * (E - U x))

def effPot (V : ℝ → ℝ) (M x : ℝ) : ℝ := x ^ 2 / 2 + V (M / x)
def angMom (V : ℝ → ℝ) (r : ℝ) : ℝ := √(r ^ 3 * deriv V r)
```

The integrand is unbounded at the turning points, so Φ is a genuinely singular (Lebesgue) integral. We do not regularize it in the definition.

3 Main theorem

Theorem 1 (Bertrand, part I)**.** *Let $0 < a < b$ and let V be C^3 on (a, b) with $V' > 0$ and $3V' + rV'' > 0$ there (every circular orbit of radius in (a, b) exists and is stable). Suppose that for every $r \in (a, b)$, every orbit with the angular momentum of the circular orbit of radius r and energy slightly above it is closed, that is, $\Phi(E) \in \pi\mathbb{Q}$. Then there is a rational $\beta > 0$ with $3 + rV''/V' = \beta^2$ on (a, b), and $V'(r) = C\, r^{\beta^2 - 3}$ for some $C > 0$.*

```
theorem forceLaw_of_closed_nearCircular {V : ℝ → ℝ} {a b : ℝ} (ha : 0 < a) (hab : a < b)
    (hV : ContDiffOn ℝ 3 V (Ioo a b))
    (hpos : ∀ r ∈ Ioo a b, 0 < deriv V r)
    (hstab : ∀ r ∈ Ioo a b, 0 < 3 * deriv V r + r * iteratedDeriv 2 V r)
    (hclosed : ∀ r ∈ Ioo a b, ∀ᶠ E in 𝓝[>] (circularEnergy V r),
      ∃ q : ℚ, orbitApsidalAngle V a b r E = q * π) :
    ∃ β : ℚ, 0 < β ∧ (∃ C > 0, ∀ r ∈ Ioo a b, deriv V r = C * r ^ ((β : ℝ) ^ 2 - 3)) ∧
      ∀ r ∈ Ioo a b, 3 + r * iteratedDeriv 2 V r / deriv V r = (β : ℝ) ^ 2
```

The orbit's allowed region is taken inside the radii (a, b), i.e. $\Omega = (M/b, M/a)$ in the x variable.

Proof outline. Fix r and the well U of its circular orbit.

1. *Morse chart* (§4). There is a C^1 map X with $X(0) = x_0$, $X' > 0$ and $U(X(s)) = U(x_0) + s^2/2$ near 0, with $X'(0) = 1/\sqrt{U''(x_0)}$.

2. *Regularization* (§5). For $0 < \varepsilon$ small, $\Phi(U(x_0) + \varepsilon) = \int_{-\pi/2}^{\pi/2} X'(\sqrt{2\varepsilon}\sin\theta)\, d\theta =: F(\varepsilon)$. The function F is continuous on $[0, \rho^2/2)$, with $F(0) = \pi X'(0) = \pi/\sqrt{U''(x_0)}$.

3. *Rationality.* By hypothesis $F(\varepsilon)/\pi \in \mathbb{Q}$ for small $\varepsilon > 0$. A continuous rational-valued function on an interval is constant, since otherwise its image would contain an irrational number. So F is constant near 0^+, and by continuity $F(0) \in \pi\mathbb{Q}$. Hence $\sqrt{U''(x_0)} \in \mathbb{Q}$.

4. *Power law.* By (1), $r \mapsto \sqrt{3 + rV''/V'}$ is continuous on (a, b) and rational-valued, hence a constant β. Then $rV'' = (\beta^2 - 3)V'$, so $(V'(r)\, r^{3-\beta^2})' = 0$ and $V' = C\, r^{\beta^2 - 3}$.

Step 3 is where continuity of Φ *in the energy* is indispensable. A function with values in $\pi\mathbb{Q}$ can perfectly well have an irrational limit. This is why the near-circular formula $\Phi \to \pi/\sqrt{U''}$, which textbooks derive by linearization, does not by itself prove the step.

Theorem 2 (Bertrand)**.** *Under the hypotheses of Theorem 1, with V smooth (C^∞) on (a, b), there is $C > 0$ such that either $V'(r) = C/r^2$ on (a, b) (inverse-square law) or $V'(r) = Cr$ on (a, b) (Hooke's law).*

```
theorem bertrand {V : ℝ → ℝ} {a b : ℝ} (ha : 0 < a) (hab : a < b)
    (hV : ContDiffOn ℝ ∞ V (Ioo a b))
    (hpos : ∀ r ∈ Ioo a b, 0 < deriv V r)
    (hstab : ∀ r ∈ Ioo a b, 0 < 3 * deriv V r + r * iteratedDeriv 2 V r)
    (hclosed : ∀ r ∈ Ioo a b, ∀ᶠ E in 𝓝[>] (circularEnergy V r),
      ∃ q : ℚ, orbitApsidalAngle V a b r E = q * π) :
    ∃ C > 0, (∀ r ∈ Ioo a b, deriv V r = C / r ^ 2) ∨ (∀ r ∈ Ioo a b, deriv V r = C * r)
```

Proof outline of Theorem 2. By the proof of Theorem 1, $V' = Cr^p$ with $p = \beta^2 - 3$, and near each circular orbit the apsidal angle is *constant*, $\Phi(U_0 + \varepsilon) = \pi/\sqrt{U''(x_0)}$. Take a *smooth* Morse chart X (§6). Then $\int_{-\pi/2}^{\pi/2} X'(\sqrt{2\varepsilon}\sin\theta)\, d\theta = \pi X'(0)$ for all small ε, which forces $X'''(0) = 0$ (§7). Differentiating $U(X(s)) = U_0 + s^2/2$ four times turns this into $5U'''^2 = 3U''U''''$ at x_0 (§8). For the power law, $U'' = p + 3$, $U''' = -(p+2)(p+3)/x_0$ and $U'''' = (p+2)(p+3)(p+4)/x_0^2$, so the condition becomes $(p+2)(p+3)^2(p-1) = 0$. Since $p + 3 = \beta^2 > 0$, we get $p = -2$ or $p = 1$.

4 The one-dimensional Morse lemma

```
theorem exists_morseChart {U : ℝ → ℝ} {Ω : Set ℝ} {x₀ : ℝ} (hΩ : IsOpen Ω) (hx₀ : x₀ ∈ Ω)
    (hU : ContDiffOn ℝ 3 U Ω) (h1 : deriv U x₀ = 0) (h2 : 0 < iteratedDeriv 2 U x₀) :
    ∃ c : MorseChart U Ω x₀, deriv c.X 0 = 1 / √(iteratedDeriv 2 U x₀)
```

The usual proof writes $U(x) - U(x_0) = (x - x_0)^2 g(x)$ with $g(x) = \int_0^1 (1-t)U''(x_0 + t(x - x_0))\, dt$, and needs differentiation under the integral sign. The formal proof avoids this. It sets $q(x) = (U(x) - U(x_0))/(x - x_0)^2$ and $s(x) = (x - x_0)\sqrt{2q(x)}$, and uses Mathlib's L'Hôpital rule (`lhopital_zero_nhdsNE`) to get $q(x) \to U''(x_0)/2$. Hence $s'(x_0) = \sqrt{U''(x_0)}$. Away from x_0, differentiating $s^2 = 2(U - U(x_0))$ gives $s' = U'/s$, which tends to $\sqrt{U''(x_0)}$ as well. So s is C^1 with positive derivative near x_0, and the local inverse given by Mathlib's inverse function theorem (`HasStrictFDerivAt.toOpenPartialHomeomorph`) is the chart X.

5 The apsidal angle in a Morse chart

```
theorem MorseChart.apsidalAngle_eq (c : MorseChart U Ω x₀) {ε : ℝ} (hε : 0 < ε)
    (hερ : ε < c.ρ ^ 2 / 2) :
    apsidalAngle U Ω x₀ (U x₀ + ε) =
      ∫ θ in -(π / 2)..π / 2, deriv c.X (√(2 * ε) * sin θ)
```

Let $\sigma = \sqrt{2\varepsilon}$. First, the allowed component equals $X((-\sigma, \sigma))$. This image is connected, lies in $\{U < E\}$ and contains x_0. Conversely, the component is an interval around x_0 that cannot contain the turning points $X(\pm\sigma)$, where $U = E$. Second, the map $\theta \mapsto X(\sigma \sin\theta)$ is a C^1 bijection from $(-\pi/2, \pi/2)$ onto that component. On it $2(E - U) = \sigma^2 \cos^2\theta$, so the Jacobian $X'(\sigma\sin\theta)\sigma\cos\theta$ cancels the singularity exactly. The change of variables uses Mathlib's Lebesgue version `integral_image_eq_integral_abs_deriv_smul`, which requires neither continuity nor integrability of the singular integrand. Continuity of $F(\varepsilon) = \int X'(\sqrt{2\varepsilon}\sin\theta)\,d\theta$ on $[0, \rho^2/2)$ follows from continuity of parametric integrals with jointly continuous integrand.

6 A smooth Morse chart

The second-order argument needs $X \in C^3$, so the C^1 chart of §4 is not enough. In §4, $q(x) = (U(x) - U(x_0))/(x - x_0)^2$ takes Lean's junk value 0 at x_0, which is harmless for C^1 but useless for higher smoothness. The smooth version instead uses

$$q = \mathrm{dslope}(\mathrm{dslope}\, U\, x_0)\, x_0,$$

and proves a real *Hadamard lemma*: if f is smooth then so is $\mathrm{dslope}\, f\, a$. Mathlib has this only for complex-differentiable functions. The proof writes $\mathrm{dslope}\, f\, a\, (x) = \int_0^1 f'(a + t(x - a))\,dt$ and shows that $x \mapsto \int_0^1 H(x,t)\,dt$ is smooth for smooth H, by induction on the order of differentiability with differentiation under the integral sign. A bump function first replaces U by a globally smooth function that agrees with it near x_0. The inverse of $s = (x - x_0)\sqrt{2q}$ is then smooth by Mathlib's `OpenPartialHomeomorph.contDiffAt_symm`.

```
theorem exists_smoothMorseChart (hΩ : IsOpen Ω) (hx₀ : x₀ ∈ Ω)
    (hU : ContDiffOn ℝ ∞ U Ω) (h1 : deriv U x₀ = 0) (h2 : 0 < iteratedDeriv 2 U x₀) :
    ∃ c : MorseChart U Ω x₀, ContDiffOn ℝ ∞ c.X (Ioo (-c.ρ) c.ρ) ∧
      deriv c.X 0 = 1 / √(iteratedDeriv 2 U x₀)
```

7 A constant apsidal angle forces $X'''(0) = 0$

Write $g = X'$. Taylor's theorem with Peano remainder (`taylor_isLittleO`) gives $g(y) = g(0) + g'(0)y + \frac{1}{2}g''(0)y^2 + R(y)$ with $|R(y)| \le \eta y^2$ for small $|y|$. Integrating along $y = \sigma\sin\theta$ and using $\int \sin\theta = 0$ and $\int \sin^2\theta = \pi/2$ over $[-\pi/2, \pi/2]$:

$$\int_{-\pi/2}^{\pi/2} X'(\sigma\sin\theta)\,d\theta = \pi X'(0) + \frac{\pi}{4} X'''(0)\,\sigma^2 + O(\eta\sigma^2).$$

If the left side equals $\pi X'(0)$ for all small σ, choosing $\eta = |X'''(0)|/8$ gives a contradiction unless $X'''(0) = 0$.

```
theorem iteratedDeriv_three_eq_zero_of_integral_const {X : ℝ → ℝ} {ρ ε₁ : ℝ} (hρ : 0 < ρ)
    (hε₁ : 0 < ε₁) (hX : ContDiffOn ℝ ∞ X (Ioo (-ρ) ρ))
    (h : ∀ ε ∈ Ioo 0 ε₁, ∫ θ in -(π / 2)..π / 2, deriv X (√(2 * ε) * sin θ) = π * deriv X 0) :
    iteratedDeriv 3 X 0 = 0
```

8 The isochrony condition

Differentiating $U(X(s)) = U(x_0) + s^2/2$ gives, with $u_k = U^{(k)}(x_0)$ and derivatives of X at 0: $u_2 X'^2 = 1$, then $X'' = -u_3 X'^2/(3u_2)$, and in general $X''' = (5u_3^2 - 3u_2 u_4)/(12u_2^4 X')$. The formal proof derives identities (1)–(4) as equalities of functions on the chart domain, each by differentiating the previous one, and evaluates the fourth at $s = 0$. Only the consequence $X'''(0) = 0 \Rightarrow 5u_3^2 = 3u_2 u_4$ is stated. This is the well-known second-order condition for an oscillator's period to be independent of its amplitude.

```
theorem MorseChart.isochrony_of_iteratedDeriv_three_eq_zero (c : MorseChart U Ω x₀)
    (hΩ : IsOpen Ω) (hU : ContDiffOn ℝ ∞ U Ω) (hX : ContDiffOn ℝ ∞ c.X (Ioo (-c.ρ) c.ρ))
    (h3 : iteratedDeriv 3 c.X 0 = 0) :
    5 * iteratedDeriv 3 U x₀ ^ 2 = 3 * iteratedDeriv 2 U x₀ * iteratedDeriv 4 U x₀
```

For a power law $V' = Cr^p$, the derivative of the effective potential near x_0 is $U'(x) = x - Kx^{-(p+2)}$ with $K = CM^{p+1}$, and $Kx_0^{-(p+2)} = x_0$ at the circular orbit. The higher derivatives follow by differentiating real powers (lemma `iteratedDeriv_effPot_powerLaw`).

9 Discussion

Size and checking. The development consists of ten files, about 1500 lines in all. It has no `sorry`, and `#print axioms Bertrand.bertrand` reports only `propext`, `Classical.choice` and `Quot.sound`. The proof was developed in parallel. For each stage we first fixed a small interface (the definition of `apsidalAngle`, the `MorseChart` structure and the exact statements of the analytic lemmas). Sub-agents then proved the Morse lemmas, the chart computation, the Taylor argument and the isochrony identity against it, while the physics layer and the assembly were written alongside. Every agent result was re-checked by a full build.

Why the local route. The textbook stage (II) compares the apsidal angle of a power law with its limits as $E \to \infty$ or $E \to 0^-$, which are $\pi/2$ and π/β^2 respectively. That needs closedness of *all* bounded orbits, a Morse chart covering the whole well and two asymptotic evaluations of singular integrals. The second-order route needs only near-circular orbits and finite Taylor expansions. It gives the stronger local theorem and reuses the same regularized integral as stage (I).

Limitations.

- *Closed orbits.* We take "closed" to mean $\Phi \in \pi\mathbb{Q}$. Deriving the apsidal-angle description from Newton's equations (Binet's equation and periodicity of the radial motion) would connect the result to Physlib's central-force library.
- *Converse.* We do not formalize the converse (that Newton and Hooke potentials satisfy the hypotheses, with $\Phi \equiv \pi$ and $\Phi \equiv \pi/2$). We checked it numerically for our definition of Φ; it also confirms that the hypotheses are not vacuous.
- *Smoothness.* We assume $V \in C^\infty$; C^5 would suffice.

Author contributions. N.L. chose the problem, directed the work and reviewed the results. Claude, an AI system developed by Anthropic, designed the formal statements and proof architecture, wrote the Lean proofs (delegating the analytic lemmas to sub-agents) and drafted the manuscript. All statements were checked by the Lean kernel. As an AI system cannot take responsibility for a publication, N.L. is accountable for its content.

References

[1] V. I. Arnold. *Mathematical Methods of Classical Mechanics*, volume 60 of *Graduate Texts in Mathematics*. Springer, 2nd edition, 1989.

[2] Joseph Bertrand. Théorème relatif au mouvement d'un point attiré vers un centre fixe. *Comptes Rendus de l'Académie des Sciences de Paris*, 77:849–853, 1873.

[3] Siu A. Chin. A truly elementary proof of Bertrand's theorem, 2014.

[4] Jacques Féjoz and Laurent Kaczmarek. Sur le théorème de Bertrand (d'après Michael Herman). *Ergodic Theory and Dynamical Systems*, 24, 2004.

[5] Herbert Goldstein, Charles Poole, and John Safko. *Classical Mechanics*. Addison-Wesley, 3rd edition, 2002.

[6] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.